

## PORT HEDLAND, Australia

WMO: 943120

Lat: 20.3725S

Lon: 118.6317E

Elev: 8

StdP: 101.22

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 10.2       | 11.6 | -10.9                         | 1.5 | 26.2 | -8.0 | 1.9 | 24.9 | 11.1                  | 21.2 | 10.4 | 22.3 | 3.7                   | 150  | 0.507 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 9.1                    | 40.9            | 21.3 | 39.2 | 21.8 | 38.0 | 22.2 | 28.4                | 33.3 | 27.9 | 32.3 | 27.4 | 31.9 | 6.4                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.2                            | 23.1 | 30.4 | 26.8 | 22.5 | 30.3 | 26.2 | 21.6 | 29.9 | 92.2          | 33.5 | 89.4 | 32.4 | 87.2 | 32.0 | 36.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |      |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|------|
| 1%                | 2.5% | 5%  | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |      |
|                   |      |     | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |      |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |
| 11.1              | 10.1 | 9.3 | DB                         | 7.8 | 44.4               | 1.5 | 1.8                                                | 6.7 | 45.7       | 5.8 | 46.7       | 4.9 | 47.7       | 3.8 | 49.0 |
|                   |      |     | WB                         | 3.5 | 30.7               | 1.0 | 2.7                                                | 2.8 | 32.7       | 2.2 | 34.2       | 1.6 | 35.8       | 0.8 | 37.7 |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 26.5  | 30.8  | 30.7  | 30.7  | 28.6  | 24.3  | 21.0  | 20.3  | 21.5  | 24.0  | 27.5  | 29.1  | 30.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 4.51  | 2.00  | 2.05  | 2.06  | 2.00  | 2.35  | 2.48  | 2.17  | 1.92  | 2.17  | 2.78  | 2.39  | 2.09 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 13    | 0     | 0     | 0     | 0     | 0     | 5     | 7     | 1     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD10.0     | 6041  | 644   | 579   | 641   | 557   | 443   | 332   | 321   | 357   | 419   | 543   | 573   | 633  | (10) |
| (11) |                                                                                  | CDD18.3     | 3012  | 386   | 346   | 382   | 307   | 184   | 87    | 68    | 100   | 169   | 284   | 323   | 374  | (11) |
| (12) |                                                                                  | CDH23.3     | 36919 | 5230  | 4651  | 5120  | 3622  | 1706  | 663   | 633   | 1061  | 1822  | 3375  | 4048  | 4989 | (12) |
| (13) | CDH26.7                                                                          | 19358       | 2869  | 2533  | 2858  | 1938  | 737   | 157   | 137   | 348   | 874   | 1889  | 2267  | 2751  | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 5.0   | 5.6   | 5.2   | 4.8   | 4.2   | 4.5   | 4.7   | 4.5   | 4.6   | 4.7   | 5.2   | 5.6   | 5.8  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 277   | 59    | 101   | 40    | 14    | 24    | 22    | 10    | 4     | 1     | 1     | 4     | 3    | (15) |
| (16) |                                                                                  | PrecMax     | 569   | 454   | 360   | 231   | 107   | 157   | 125   | 75    | 41    | 9     | 8     | 32    | 38   | (16) |
| (17) |                                                                                  | PrecMin     | 57    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 133   | 88    | 90    | 59    | 25    | 39    | 33    | 19    | 10    | 2     | 2     | 5     | 9    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 41.9  | 41.9  | 41.3  | 39.2  | 35.3  | 31.8  | 31.2  | 33.2  | 37.4  | 41.6  | 42.2  | 42.2 | (19) |
| (20) |                                                                                  |             | MCWB  | 22.1  | 23.3  | 21.7  | 20.9  | 18.3  | 18.6  | 16.4  | 16.9  | 17.6  | 18.8  | 19.5  | 20.5 | (20) |
| (21) |                                                                                  | 2%          | DB    | 39.0  | 39.0  | 39.2  | 37.2  | 33.8  | 29.9  | 29.6  | 31.2  | 35.2  | 39.2  | 39.7  | 39.4 | (21) |
| (22) |                                                                                  |             | MCWB  | 24.1  | 23.8  | 22.5  | 20.7  | 18.4  | 17.2  | 15.7  | 16.3  | 17.5  | 18.9  | 20.2  | 21.9 | (22) |
| (23) |                                                                                  | 5%          | DB    | 36.9  | 37.0  | 37.8  | 36.0  | 32.2  | 28.5  | 28.1  | 30.0  | 33.2  | 37.1  | 37.8  | 37.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 25.8  | 24.8  | 23.3  | 20.8  | 18.3  | 16.6  | 15.4  | 15.6  | 17.4  | 19.3  | 20.5  | 23.4 | (24) |
| (25) |                                                                                  | 10%         | DB    | 35.2  | 35.3  | 36.1  | 34.6  | 30.9  | 27.1  | 26.9  | 28.8  | 31.6  | 35.0  | 35.7  | 35.5 | (25) |
| (26) |                                                                                  |             | MCWB  | 26.2  | 25.8  | 24.1  | 21.1  | 17.9  | 16.0  | 15.3  | 15.2  | 17.4  | 19.7  | 21.5  | 24.2 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 29.0  | 29.2  | 28.8  | 27.5  | 25.0  | 24.3  | 22.2  | 21.6  | 23.4  | 25.7  | 27.2  | 28.1 | (27) |
| (28) |                                                                                  |             | MADB  | 34.6  | 34.3  | 33.8  | 31.7  | 28.2  | 26.6  | 24.8  | 25.2  | 27.4  | 30.8  | 31.6  | 33.8 | (28) |
| (29) |                                                                                  | 2%          | WB    | 28.2  | 28.4  | 27.8  | 26.3  | 23.6  | 22.1  | 20.6  | 20.3  | 21.8  | 24.7  | 26.1  | 27.2 | (29) |
| (30) |                                                                                  |             | MADB  | 33.1  | 33.2  | 31.8  | 29.9  | 27.1  | 24.9  | 24.2  | 24.4  | 27.4  | 29.5  | 31.2  | 32.5 | (30) |
| (31) |                                                                                  | 5%          | WB    | 27.7  | 27.8  | 27.3  | 25.6  | 22.5  | 20.9  | 19.1  | 19.3  | 21.0  | 24.0  | 25.5  | 26.7 | (31) |
| (32) |                                                                                  |             | MADB  | 32.3  | 32.2  | 31.4  | 29.5  | 26.7  | 24.3  | 23.6  | 24.4  | 26.7  | 29.1  | 30.7  | 31.9 | (32) |
| (33) |                                                                                  | 10%         | WB    | 27.2  | 27.3  | 26.8  | 24.9  | 21.5  | 19.5  | 17.9  | 18.2  | 20.3  | 23.2  | 24.8  | 26.2 | (33) |
| (34) |                                                                                  |             | MADB  | 31.9  | 31.6  | 31.0  | 29.1  | 25.9  | 23.6  | 22.7  | 23.9  | 26.0  | 28.5  | 30.1  | 31.4 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 9.1   | 9.3   | 10.6  | 12.5  | 12.8  | 12.5  | 13.8  | 15.2  | 15.6  | 14.7  | 12.9  | 10.3 | (35) |
| (36) |                                                                                  |             | MADB  | 12.5  | 12.5  | 13.6  | 14.5  | 14.7  | 13.5  | 15.1  | 16.6  | 18.1  | 17.7  | 16.5  | 13.8 | (36) |
| (37) |                                                                                  |             | MCWBR | 4.6   | 5.0   | 5.5   | 6.0   | 6.4   | 6.0   | 6.4   | 6.8   | 6.9   | 6.5   | 6.5   | 5.5  | (37) |
| (38) |                                                                                  | 5% WB       | MADB  | 9.7   | 9.9   | 10.3  | 11.0  | 11.6  | 10.0  | 11.8  | 14.1  | 15.4  | 13.5  | 11.9  | 10.0 | (38) |
| (39) |                                                                                  |             | MCWBR | 3.9   | 4.4   | 4.5   | 4.8   | 5.8   | 5.3   | 5.8   | 6.4   | 6.5   | 5.9   | 5.4   | 4.3  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.408 | 0.399 | 0.382 | 0.355 | 0.336 | 0.322 | 0.317 | 0.317 | 0.340 | 0.366 | 0.373 | 0.383 | (40) |      |
| (41) |                                                                                  | taud        | 2.475 | 2.518 | 2.549 | 2.573 | 2.549 | 2.579 | 2.569 | 2.523 | 2.474 | 2.439 | 2.461 | 2.511 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 937   | 934   | 927   | 913   | 893   | 889   | 905   | 938   | 952   | 954   | 965   | 962   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 119   | 112   | 104   | 95    | 90    | 84    | 87    | 98    | 110   | 120   | 120   | 114   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.27  | 7.00  | 6.32  | 5.63  | 4.72  | 4.28  | 4.62  | 5.63  | 6.64  | 7.49  | 7.98  | 7.88  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.52  | 0.54  | 0.52  | 0.29  | 0.37  | 0.33  | 0.29  | 0.19  | 0.27  | 0.24  | 0.21  | 0.38  | (45) |      |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |      |
|------|------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |      |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m)  |      |
| (46) | Station Only           | +0.49   | N/A    | N/A     | N/A   | +0.48 | +0.52       | N/A     | N/A     | N/A     | +158 | (46) |
| (47) | Regional (7 neighbors) | +0.46   | N/A    | N/A     | N/A   | +0.48 | +0.56       | N/A     | N/A     | N/A     | +158 | (47) |

Nomenclature: See separate page